Project Management Process

- Is a controlled process of initiating, planning, executing, and closing down a project.
- The project management process, involves four phases:

1. Initiating the project
2. Planning the project
3. Executing the project
4. Closing down the project

Representing and Scheduling Project Plans

- These planning documents can take the form of graphical or textual reports.
- The most commonly used methods are Gantt charts and network diagrams.
- The differences between Gantt charts and network diagrams :

◦ A **Gantt chart** shows the **duration of tasks**, whereas a network diagram shows the sequence dependencies between **tasks**.
◦ A Gantt chart shows the **time overlap of tasks**, whereas a network diagram does not show time overlap but does show which tasks could be done in parallel.
◦ Some forms of Gantt charts can show **slack time available within an earliest start and latest finish date.**

Representing Project Plans

- Resources: Any person, group of people, piece of equipment, or material used in accomplishing an activity.
- Critical path scheduling: A scheduling technique in which the order and duration of a sequence of **task activities directly affect the completion date of a project.**

Calculating Expected Time Durations Using PERT

- (Program Evaluation Review Technique) is a technique that uses optimistic, pessimistic, and realistic time estimates to calculate the expected time for a particular task.

$$ET = \frac{o + 4r + p}{6}$$

- where

ET = expected time for the completion for an activity
o = optimistic completion time for an activity
r = realistic completion time for an activity
p = pessimistic completion time for an activity

Constructing a Gantt Chart and Network Diagram

1. Identify each activity to be completed in the project.

For example activities in project include:

- Requirements collection
- Screen design
- Report design
- Database design
- User documentation creation
- Software programming
- System testing
- System installation

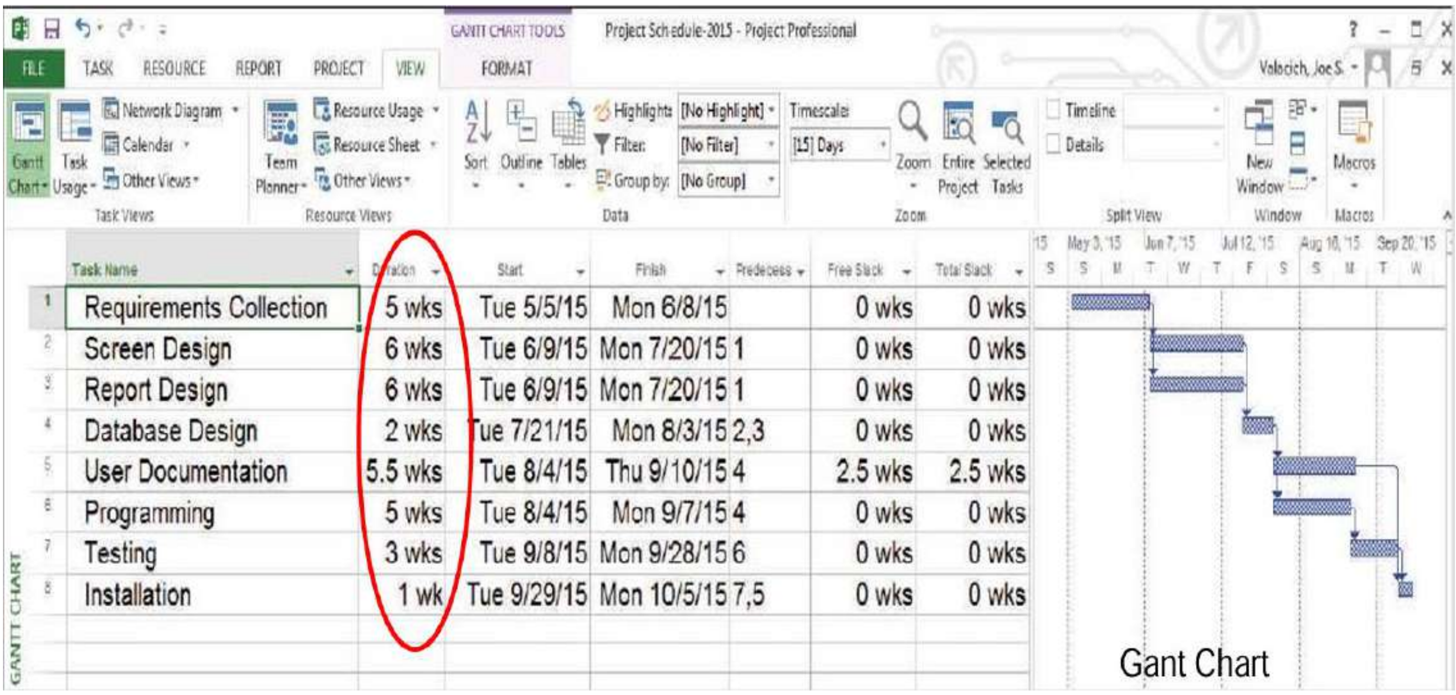
2. Determine time estimates and calculate the expected completion time for each activity.

ACTIVITY	TIME ESTIMATE (in weeks) o	r	p	EXPECTED TIME (ET) $\frac{o + 4r + p}{6}$
1. Requirements Collection	1	5	9	5
2. Screen Design	5	6	7	6
3. Report Design	3	6	9	6
4. Database Design	1	2	3	2
5. User Documentation	2	6	7	5.57
6. Programming	4	5	6	5
7. Testing	1	3	5	3
8. Installation	1	1	1	1

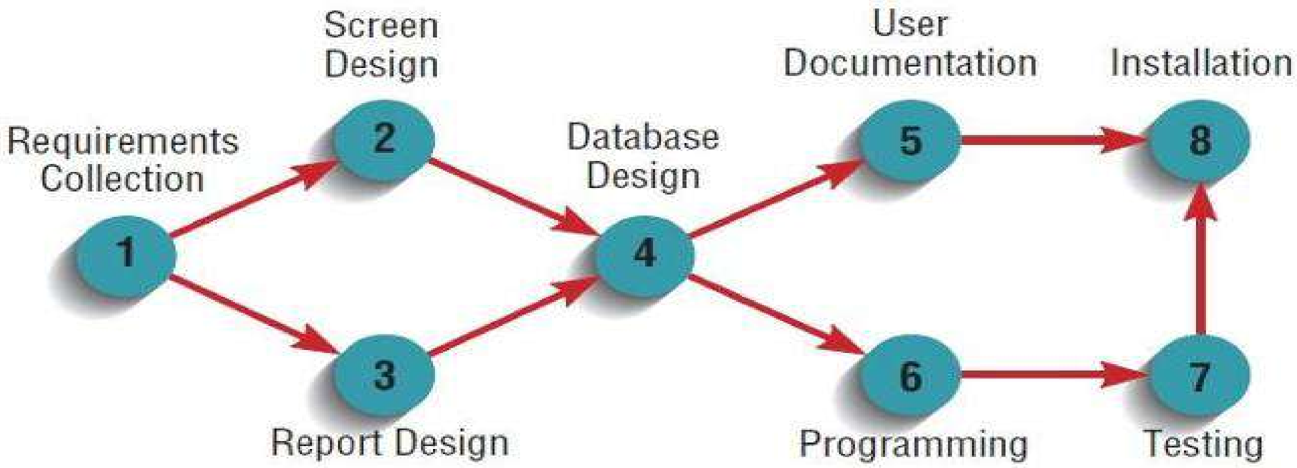
3. Determine the sequence of the activities and precedence relationships among all activities by constructing a Gantt chart and network diagram.

ACTIVITY	PRECEDING ACTIVITY
1. Requirements Collection	—
2. Screen Design	1
3. Report Design	1
4. Database Design	2,3
5. User Documentation	4
6. Programming	4
7. Testing	6
8. Installation	5,7

Sequence of Activities



Constructing a Gantt Chart and Network Diagram cont.



4.Determine the critical path: **is the shortest time in which a project can be completed.**

- Any activity on the critical path that is **delayed** in completion thus **delays the entire project.**
- Nodes (activities) not on the critical path, however, can be delayed (for some amount of time) **without delaying the final completion** of the project.

Calculating Expected time

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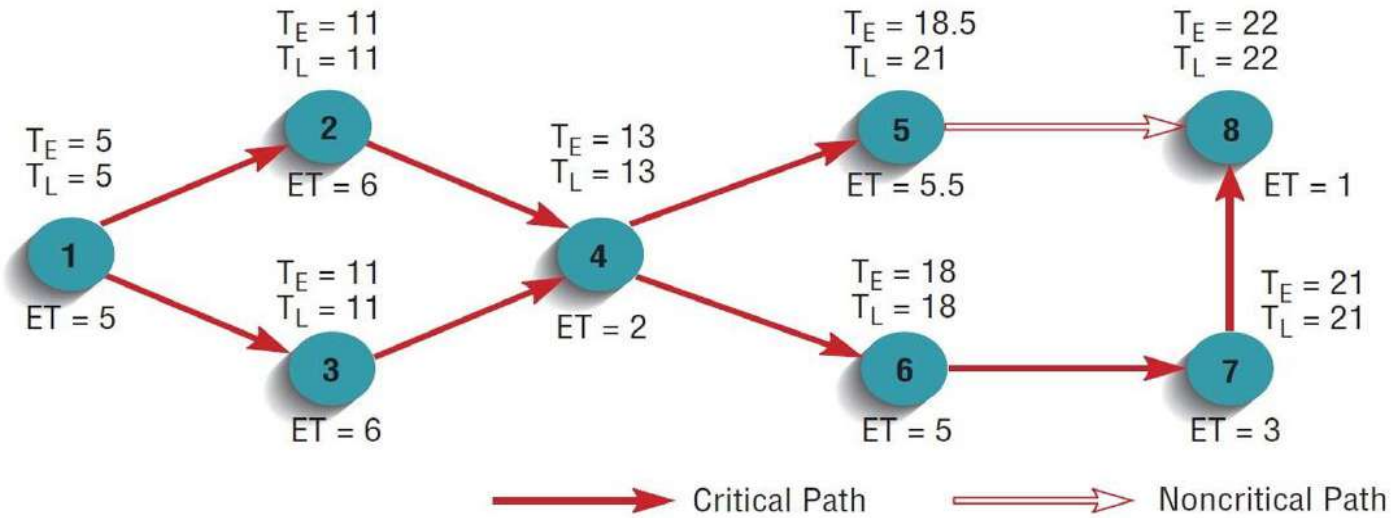
- **Slack time**: The amount of time that an activity can be delayed without delaying the project.



Constructing a Gantt Chart and Network Diagram cont.

● Activity's earliest expected completion time (TE) by summing the estimated time (ET) for each activity from left to right (i.e., in precedence order), starting at activity 1 and working toward activity 8.
● The latest expected completion time (TL) refers to the time in which an activity can be completed without delaying the project.
● If two or more activities **precede** an activity, the largest expected completion time of these activities is used in calculating the new activity's expected completion time.
● For example, because activity 8 is preceded by both activities 5 and 7, the largest expected completion time between 5 and 7 is 21, so TE for activity 8 is 21 + 1, or 22.

The following table shows:
For each activity, the slack time and whether the task is on the critical path or not.

ACTIVITY	T_E	T_L	SLACK $T_L - T_E$	ON CRITICAL PATH
1	5	5	0	✓
2	11	11	0	✓
3	11	11	0	✓
4	13	13	0	✓
5	18.5	21	2.5	
6	18	18	0	✓
7	21	21	0	✓
8	22	22	0	✓

Using Project Management Software

● A wide variety of automated project management tools are available to help you manage a development project.

● Features include the ability to define and order tasks, assign resources to tasks, and easily modify tasks and resources.

● These systems vary in the number of activities supported, the complexity of relationships, system processing and storage requirements.

● Commercial software such as Microsoft Project.

● Free software for project Management such as OpenProj or EasyProjectPlan.